Sarah Lamik

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Available for full-time positions from January 2027

ML engineer with research experience in designing and running independent experiments across reinforcement learning, graph neural networks, self-supervised learning and JAX-based implicit deep learning, with a focus on ablation-driven analysis of model behavior.

Education

CentraleSupélec, Université Paris-Saclay – ranked 13^{th} worldwide on Shanghai ranking — Sept. 2022 – Apr. 2026
*Engineering Degree, specialization in Artificial Intelligence - GPA: **4.09 / 4.33** (S1), **4.26/4.33** (S2)*

- Relevant coursework: Statistical Learning, Convex Optimization, Reinforcement Learning (MDPs, bandits, policy gradient), Graph Machine Learning, Probabilistic Modeling, SAT Solving and Combinatorial Reasoning, Game Theory, Signal Processing, Cloud Computing.

Lycée Hoche – CPGE PC* — Sept. 2020 – Jun. 2022

- Intensive two-year selective program in Mathematics and Physics.
- Advanced Mathematics, Algorithms, Complexity Theory and Theoretical Physics.

Research Experience

Research Intern – RTE & Mines Paris (Ongoing) — May 2026 – Oct. 2026
Graph Neural Networks and Implicit Deep Learning

- Co-authoring a **scientific publication** on the intersection of implicit deep learning and GNN prediction
- Designed and implemented a Deep Equilibrium (DEQ) implicit layer within an existing library in JAX, targeting fixed-point convergence guarantees.
- Extended GNN architectures to hypergraphs and multigraphs, with theoretical analysis of expressiveness and structural properties
- Investigated linear solver selection and sparsity-aware graph representations for scalable inference.

Industry AI Experience

European Central Bank — Jan. 2025 – Jun. 2025
Risk Management Department - Received an official letter of recommendation

- Developed anomaly detection tools for financial risk monitoring.
- Designed and deployed a Retrieval-Augmented Generation (RAG) assistant for internal reports. Used by 20
- Implemented transformer-based NLP pipelines on large collections of unstructured documents.

Audacia – Venture Capital — Jul. 2024 – Dec. 2024

- Conducted technical due diligence on AI startups, evaluated scientific and technological feasibility of AI-based industrial solutions.

Academic Research Projects

Autonomous Driving with Reinforcement Learning — 2026 - GitHub

- Implemented and compared DQN, Double-DQN and Prioritized Experience Replay.
- Performed behavioral and performance analyses of trained agents.
- Evaluated the impact of algorithmic choices through ablation studies.

Explainable Resource Allocation and Data Disclosure — 2026 - GitLab

- Studied the privacy risks induced by explanations generated in SAT-based resource allocation systems.
- Formalized contrastive explanations using Minimum Unsatisfiable Sets (MUS).
- Designed quantitative metrics for measuring information disclosure.
- Investigated the trade-off between explainability and privacy under rank-k envy-freeness constraints.
- Tools: PySAT, SAT solving, MUS extraction.

Explainability in Linear Optimization
Reference letter by Vincent Mousseau (MICS, CentraleSupélec) — 2026 - GitHub

- Developed algorithms for generating (1-1), (1-m), (m-1) and mixed explanations in optimization problems.
- Conducted large-scale experiments on transportation and healthcare datasets.
- Studied the existence conditions of optimization-based explanations.
- Tools: Gurobi, scikit-learn, NetworkX.

Graph Neural Networks: GCN vs GAT Ablation Study 2026 - Github

- Investigated the architectural trade-offs between GCN and GAT for node classification and link prediction on large-scale graphs (git_web_ml: 37K nodes, 290K edges; WikiOGB)
- Implemented hard negative sampling to address the false negative issue in link prediction evaluation.
- Best model (3-layer GCN + MLP decoder) achieved 93.3% link prediction accuracy; best classification model (2-layer GAT) reached 78.3% accuracy.
- Tools: PyTorch Geometric, scikit-learn, Node2Vec baseline.

Technical Skills

Deep Learning & ML: PyTorch, PyTorch Geometric, JAX, TensorFlow, scikit-learn, NumPy, pandas
Optimization & Symbolic AI: Gurobi, PySAT, Constraint Programming, SAT Solving
Infrastructure : Git, Linux, SQL, AWS, CI/CD
Web/App: React, Node.js, Streamlit

Languages

French (native), English (fluent, C1), German (intermediate, B1)